MILESTONE 1

EDA & Preprocessing Report

Sales Forecasting and Demand Prediction

AI & Data Science Track

July 10, 2026

Dataset: Rossmann Store Sales (Kaggle) | Deliverable: cleaned_rossmann_data.csv (844,392 × 38)

Abstract

This report documents the data collection, exploratory data analysis (EDA), and preprocessing pipeline executed for Milestone 1 of the Sales Forecasting and Demand Prediction project. It summarizes the dataset structure, data quality findings, outlier treatment strategy, skewness analysis, feature engineering decisions, and the visual patterns uncovered during EDA. It also flags one open scoping question — the absence of product-level granularity in the source dataset — that should be resolved before proceeding to model development.

1. Executive Summary

The dataset used for this milestone is the Rossmann Store Sales dataset, sourced from Kaggle, containing daily historical sales records for 1,115 stores. After merging the transactional and store-attribute files, the working dataset contained 1,017,209 rows across 18 columns. No duplicate rows were found in either source file.

The pipeline covers: data quality inspection, missing-value imputation, closed-store filtering, IQR-based outlier detection and capping, time-based and promotional feature engineering (including two corrected logical bugs from an earlier draft), categorical encoding, feature scaling, and a structured set of dark-themed visualizations. The final cleaned and encoded dataset contains 844,392 rows across 38 columns and has been exported as cleaned_rossmann_data.csv.

***Key Finding**. The dataset is store-level, not product-level. This has direct implications for scope and is discussed in Section 2.*

2. Dataset Description and a Critical Scoping Question

2.1 Source and Structure

The data originates from the Kaggle Rossmann Store Sales competition and is split into train.csv (historical sales with the target variable) and test.csv (held-out rows without the target, matching the standard Kaggle train/test convention), plus a store.csv file with static store attributes.

The granularity of every row is: one Store × one Date.

Available fields describe the store (type, assortment level, competition distance, promotions) and the day (day of week, holiday flags, promo status) — not any individual product, SKU, or item category.

2.2 Why This Matters

The Milestone 1 brief requests, among other things, product details and product categories as part of the feature set. The Rossmann dataset, as currently scoped, does not contain product-level data of any kind — there is no product ID, product name, category, or per-item price anywhere in the source files. This is a structural property of the dataset, not a gap introduced during preprocessing.

Decision Required

There are two valid ways to proceed, and the choice affects how the forecasting objective should be worded in later milestones:

- Option A — Keep the current dataset. Reframe the forecasting target as aggregate daily store-level demand rather than product-level demand. This is a legitimate and common formulation of demand forecasting in retail analytics, and everything completed so far in this milestone remains valid under this framing.

- Option B — Switch datasets. If product-level forecasting is a hard requirement, a different source dataset containing SKU/product identifiers (e.g. other Kaggle retail-forecasting datasets, or a UCI dataset with item-level transactions) would be needed, and the preprocessing pipeline would need to be substantially reworked around a product dimension.

This report and the accompanying notebook proceed under Option A, since no product-level requirement has been confirmed. This assumption should be validated with the track supervisor before Milestone 2 begins.

2.3 Other Fields Requested but Not Present

For completeness, two additional fields mentioned in the brief are only partially available:

- Customer information: Only an aggregate daily Customers count per store is available — no demographic or individual-level customer data exists in this dataset.
- Weather: Not present in any of the source files. Holiday effects (StateHoliday, SchoolHoliday) are available and were used; weather would require joining an external dataset by date and store region, which was out of scope for this milestone.

3. Data Collection and Quality Inspection

- Source files: train.csv (sales transactions), store.csv (store attributes), joined via a left merge on Store.
- Duplicate rows: 0 in train.csv, 0 in store.csv.
- Merged shape: 1,017,209 rows × 18 columns.

4. Initial Exploratory Data Analysis

4.1 Summary Statistics

The minimum value of 0 for both Sales and Customers corresponds to days on which a store was closed — this is addressed as a structural filtering step in Section 5.2, not treated as a statistical outlier.

Statistic	Sales	Customers
Count	1,017,209	1,017,209
Mean	5,773.82	633.15
Std. Dev.	3,849.93	464.41
Minimum	0	0
25th Percentile	3,727	405
Median (50th)	5,744	609
75th Percentile	7,856	837
Maximum	41,551	7,388

Table 1 — Descriptive statistics for the target variable and its closest correlate, computed on the full merged dataset (including closed-store days).

4.2 Missing Value Analysis

Column	Missing Count	% of Rows
CompetitionDistance	2,642	0.26%
CompetitionOpenSinceMonth	323,348	31.79%
CompetitionOpenSinceYear	323,348	31.79%
Promo2SinceWeek	508,031	49.94%
Promo2SinceYear	508,031	49.94%
PromoInterval	508,031	49.94%

Table 2 — Columns with missing values in the merged dataset, out of 1,017,209 rows.

Key Finding*. The Promo2-related fields are missing for essentially half the dataset because Promo2 is a binary flag — these fields are only populated for stores where Promo2 = 1. Missingness here is structural (not applicable), not random, and was handled accordingly (see Section 5.2).*

5. Outlier Detection and Treatment

5.1 Detection (IQR Method)

Outliers were quantified using the interquartile range rule: a value is flagged if it falls outside [Q1 – 1.5×IQR, Q3 + 1.5×IQR]. The figure below shows the raw distributions of the three key numerical columns before any treatment.

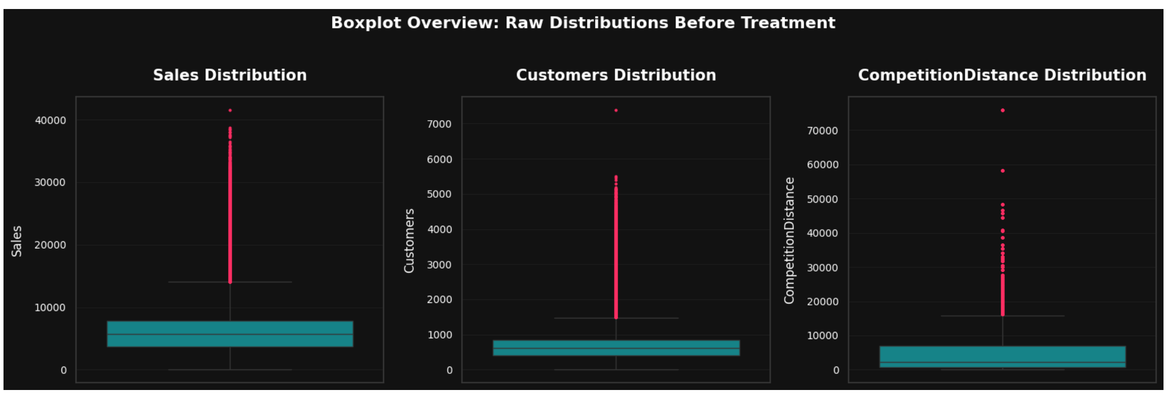

Figure 1 — Boxplot overview of Sales, Customers, and CompetitionDistance before any outlier treatment. All three show a long right tail, consistent with retail sales data where most values cluster at moderate levels with a smaller number of unusually high-performing days or high-competition-distance stores.

5.2 Treatment Strategy and Rationale

Two different treatments were applied depending on the role of each column:

- Closed-store rows (Open = 0): removed entirely (844,392 rows retained out of 1,017,209). These are not statistical outliers but a structural business condition — a closed store cannot have organic sales, so including these zero-sales days would distort both the outlier analysis and any downstream model.
- Customers and CompetitionDistance: treated with IQR-based capping (winsorization) rather than row removal, so that every remaining row is preserved — this matters because the lag

features built in Section 7 depend on having a continuous per-store time series with no gaps introduced by outlier removal.

- Sales: intentionally not capped. As the target variable, its high values (holiday peaks, promotional spikes) represent genuine business events rather than data-entry errors; capping the target would erase signal the forecasting model needs to learn.

Column	Values Capped	Capped Range
Customers	40,853 (4.84%)	[0, 1,454]
CompetitionDistance	83,041 (9.83%)	[0, 16,135]

Table 3 — IQR capping results, computed after closed-store rows were removed (n = 844,392).

Note: the lower bound for both columns is floored at 0, since neither a customer count nor a physical distance can be negative — the raw IQR formula would otherwise produce a mathematically negative lower bound with no real-world meaning.

6. Skewness Analysis

Skewness was measured before any treatment and again after cleaning and feature engineering, using the rule of thumb that |skew| < 0.5 is approximately symmetric, 0.5–1 is moderate, and > 1 is high.

Column	Skew (Before)	Skew (After)	Interpretation
Sales	0.641	1.594	See note below
Customers	1.599	0.803	Improved by capping
CompetitionDistance	2.929	1.212	Improved by capping
Sales (log1p)	—	-0.638	Near-symmetric
Sales_Lag_7	—	1.120	New feature, moderately skewed
CompetitionOpen	—	7.548	Very high (see note below)

Table 4 — Skewness before treatment (on the full merged dataset, including closed-store days) versus after treatment (post-filtering, imputation, and IQR capping).

Key Finding. *Sales skewness increased after cleaning (0.641 → 1.594). This is expected, not an error: removing closed-store days eliminates the large clump of zero-sales rows that was previously pulling the distribution toward the center. The remaining distribution — sales on days the store was actually open — has a genuinely longer right tail, which the log(1+x) transform (skew −0.638) brings much closer to symmetric. This transformed column (Sales_Log) is retained for visualization and potential modeling use, without overwriting the original target.*

Key Finding. *CompetitionOpen shows very high skew (7.548) because a large share of rows have no recorded competition-opening date and are correctly assigned a value of 0 (see the bug fix in Section 7), creating a large spike at zero next to a long tail of stores with decades of competition exposure. This is a known and explainable property of the feature, not a processing error.*

7. Feature Engineering

The following features were engineered from the raw and cleaned data:

- Time-based features: Year, Month, Day, WeekOfYear, DayOfYear, Quarter, derived directly from Date.

- Competition exposure: CompetitionOpen, the number of months a competitor has been open, relative to the current row's date.
- Promotional flags: Promo (existing), Promo2Active (derived — whether the store's long-term promotion is active in the current calendar month).
- Lag features: Sales_Lag_7 and Sales_Lag_30, representing the same store's sales exactly 7 and 30 calendar days prior, to give the model access to recent historical demand.
- Encoding: StateHoliday mapped to binary; StoreType, Assortment, DayOfWeek one-hot encoded.
- Scaling: CompetitionDistance, Sales_Lag_7, and Sales_Lag_30 standardized via StandardScaler.

7.1 Corrections Made to the Original Feature Logic

During review, three logical issues were identified and corrected before this version of the pipeline:

- CompetitionOpen: the original formula computed months-since-competition-opened using CompetitionOpenSinceYear/Month, which had been imputed to 0 for missing rows. Without a guard, this produced an inflated, meaningless value implying competition had existed since year 0. A CompetitionData_Missing flag is now captured before imputation and used to force CompetitionOpen = 0 for those rows.
- Promo2Active: the original logic compared the numeric month (e.g. 7) against PromoInterval, which stores three-letter month abbreviations (e.g. "Jan,Apr,Jul,Oct"). Since a number can never match a name, this feature always evaluated to 0. The month is now converted to its abbreviation before the comparison.
- Sales_Lag_7: the original implementation used groupby(...).shift(7), which offsets by row position, not by calendar time. Because closed-store rows are removed before this step, a 7-row shift does not reliably correspond to a 7-calendar-day lag whenever closed days fall in between. This was replaced with a calendar-date self-join, and the same logic was extended to add a second, 30-day lag feature.

8. Visual EDA Summary

A structured set of dark-themed, consistently styled visualizations was produced in the accompanying notebook (Milestone_1_updated.ipynb), covering:

- Average daily sales trend over time (weekly and yearly seasonality).
- Average sales on promotional versus non-promotional days.
- Sales distribution by store type and by assortment level.
- Average sales by day of week.
- Long-term promotion (Promo2) effect on average sales.
- A correlation heatmap across the core numerical features, including the corrected CompetitionOpen and Promo2Active features and both lag features.

Note on this report: the numerical results in Sections 4–6 are taken directly from the executed notebook outputs supplied for this report. The remaining chart outputs listed above (trend line, promo/store-type/assortment/day-of-week bar and box plots, and the heatmap) were not yet supplied at the time of *writing; their specific values and correlation coefficients should be read directly from the corresponding notebook cells once re-run, and can be folded into a future revision of this report on request.*

9. Key Insights for Downstream Modeling

- Store closures must be filtered, not modeled, since a closed store's zero sales is a structural condition rather than a demand signal.
- Log-transforming Sales is worth testing as a modeling target, given its post-cleaning skew of 1.594 versus −0.638 after log1p.
- Promo2SinceWeek/Year and PromoInterval missingness is structural (tied to Promo2 = 0), and was imputed accordingly rather than treated as random missing data.
- Two lag features (7-day and 30-day) are now available, giving the model both a short-term and a medium-term view of recent demand per store.
- The product-level scoping question in Section 2 should be resolved before Milestone 2, since it determines whether the forecasting target remains store-level aggregate demand or requires a different dataset entirely.